



Case Study: Retail Marketing Campaign Analysis

Business Scenario

You are working as a Data Analyst for an E-Commerce Company.

The company recently launched multiple marketing campaigns and wants to understand:

- Customer purchase behavior
- Campaign performance
- Product category performance
- Regional performance
- Probability of customer purchases
- Forecasting future campaign outcomes
- Estimating future order volumes

Your task is to clean the data, perform analysis, answer business questions, and provide statistical insights.

Download Dataset

<https://drive.google.com/file/d/1LnRWVmY92hw93jxuu4hIHCDeuRcRIKxE/view?usp=sharing>

Dataset Structure

Column	Description
Customer_ID	Unique customer identifier
Gender	Male/Female
Age_Group	18-25, 26-35, 36-45, 46+
Region	North, South, East, West
Campaign_Type	Email, SMS, Social Media
Clicked_Ad	Yes/No
Purchased	Yes/No
Product_Category	Electronics, Fashion, Grocery, Home Decor
Order_Value	Purchase amount
Order_Date	Transaction date

Intentional Data Issues

The dataset contains:

Missing Values

- Gender
- Product_Category
- Order_Value

Duplicate Records

Some customer entries are duplicated.

Inconsistent Values

Examples:

Male
male
MALE

Female
female
FEMALE

Students must clean these issues before analysis.

Part A: Data Cleaning

Q1

Load the dataset.

Display:

- First 5 rows
 - Last 5 rows
-

Q2

Check:

- Shape
 - Columns
 - Data Types
-

Q3

Identify missing values.

Questions:

1. Which columns contain missing values?
 2. How many missing values exist in each column?
-

Q4

Handle missing values.

Questions:

1. How would you handle missing values in Gender?
2. How would you handle missing values in Product_Category?
3. How would you handle missing values in Order_Value?

Justify your approach.

Q5

Identify duplicate records.

Questions:

1. Number of duplicates
 2. Remove duplicates
-

Q6

Standardize categorical values.

Convert:

male
MALE
Male

into

Male

Similarly for Female.

Part B: Exploratory Data Analysis

Q7

Find:

- Total Customers
 - Total Purchases
 - Total Revenue
-

Q8

Find purchase count by Gender.

Q9

Find purchase count by Region.

Q10

Find purchase count by Product Category.

Q11

Find average Order Value by Product Category.

Q12

Find average Order Value by Region.

Part C: GroupBy Analysis

Q13

Using GroupBy:

Find:

Revenue by Region.

Q14

Using GroupBy:

Find:

Revenue by Campaign Type.

Q15

Using GroupBy:

Find:

Average Order Value by Gender.

Q16

Find:

Purchases by:

Region and Product Category.

Part D: Crosstab Analysis

Q17

Create Crosstab:

Gender vs Purchased

Questions:

- Which gender purchases more?
 - Compare purchase percentages.
-

Q18

Create Crosstab:

Campaign Type vs Purchased

Questions:

- Which campaign performs best?
 - Which campaign performs worst?
-

Q19

Create Crosstab:

Region vs Purchased

Interpret the results.

Part E: Pivot Tables

Q20

Create Pivot Table:

Rows = Region

Values = Order_Value

Aggregation = Sum

Q21

Create Pivot Table:

Rows = Product_Category

Columns = Gender

Values = Order_Value

Aggregation = Mean

Q22

Create Pivot Table:

Rows = Campaign_Type

Columns = Purchased

Values = Customer_ID

Aggregation = Count

Part F: Probability Analysis

Q23: Marginal Probability

Find:

Probability that a randomly selected customer made a purchase.

Q24: Marginal Probability

Find:

Probability that a randomly selected customer belongs to the South Region.

Q25: Joint Probability

Find:

Probability that a customer:

- Purchased
AND
 - Belongs to South Region
-

Q26: Joint Probability

Find:

Probability that a customer:

- Purchased
AND

- Clicked the Ad
-

Q27: Conditional Probability

Find:

Probability that a customer purchased given that they clicked the ad.

Q28: Conditional Probability

Find:

Probability that a customer purchased given that they belong to the South Region.

Q29

Interpret all probability results from a business perspective.

Part G: Bernoulli Distribution

Q30

Create binary column:

Purchased_Binary

Yes = 1

No = 0

Q31

Find:

Probability of Purchase.

Q32

Why is this a Bernoulli Experiment?

Explain.

Q33

If 100 new customers are targeted:

How many purchases would you expect?

Part H: Binomial Distribution

Scenario

Marketing plans to target 20 customers.

Use purchase probability obtained earlier.

Q34

Find probability of exactly:

5 purchases.

Q35

Find probability of exactly:

10 purchases.

Q36

Find probability of at least:

3 purchases.

Q37

Find expected purchases among 20 customers.

Q38

Interpret the results from a business perspective.

Part I: Poisson Distribution

Q39

Calculate:

Daily Orders.

Q40

Find:

Average Orders Per Day (λ).

Q41

Find probability of exactly:

15 orders tomorrow.

Q42

Find probability of exactly:

25 orders tomorrow.

Q43

Find probability of more than:

30 orders tomorrow.

Q44

Interpret all Poisson results.

How can Operations Teams use this information?

Final Business Report

Prepare a report answering:

1. Which region performs best?
2. Which campaign performs best?
3. Which product category generates highest revenue?
4. What is the customer purchase probability?

5. What purchase outcomes can marketing expect from future campaigns?
6. What order volumes can operations expect in the future?
7. Provide 5 business recommendations.



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